Interaction Similarity

Toward a General Theory of Comparable Interactions

Cross-disciplinary foundational research paper

Abstract

Interaction research routinely compares interfaces, traces, users, policies, models, and outcomes using the language of sameness or similarity. The terms are often underspecified. Formal methods show that even behaviorally oriented equivalence divides into trace, testing, simulation, bisimulation, and related relations, each preserving different observations. Psychology and pattern recognition show that similarity depends on selected features, weighting, context, asymmetry, and representation. Process mining shows that model-to-model, log-to-log, and log-to-model comparisons answer different questions. Measurement theory adds a further constraint: numerical comparisons are valid only under declared scales, observation models, uncertainty, and admissible transformations.

This paper develops a typed theory of interaction comparison. It concludes that no context-free relation of interaction similarity is scientifically defensible. Interaction token identity is historical identity under a declared boundary; interaction type identity is classification under an explicit abstraction; equivalence is exact preservation relative to a declared observation contract; similarity is graded proximity under specified dimensions and metrics; and incomparability arises when targets, boundaries, observation models, scales, or validity domains cannot be aligned without unsupported assumptions. The paper proposes a layered comparison object spanning semantics, coupling structure, protocol, opportunity, trace, behavior, policy, strategy, outcome, cause, and measurement. It leaves behind a similarity ontology, taxonomy, equivalence hierarchy, similarity contract, comparison protocol, admissible-transformation matrix, and falsification matrix.

1. Problem Statement

Two interactions may share an implementation yet differ because participants, goals, states, permissions, or histories differ. Two different implementations may realize the same protocol or semantic operation. Two traces may be identical while arising from different mechanisms, and different traces may express the same strategy or achieve the same outcome. A novice and an expert can encounter the same projection but possess different believed action sets. Accessibility variants can preserve operation semantics while changing sequence, latency, discoverability, and recovery.

The scientific problem is therefore not to ask whether two interactions are simply similar. It is to determine: similar with respect to what object, under which abstraction, for which observer and task, using what evidence, metric, tolerance, and uncertainty model? Without these declarations, similarity claims collapse distinctions established by prior work on representation, opportunity, interaction, behavior, and measurement.

2. Method

The review used a concept-centered comparative method across similarity theory, formal equivalence, graph and sequence comparison, process mining, pattern recognition, behavioral science, machine learning, and HCI. Foundational sources were prioritized: Tversky on feature-based similarity; metric and edit-distance traditions; dynamic time warping; graph edit distance; Milner and van Glabbeek on process equivalence; bisimulation and trace equivalence; van der Aalst on process mining; and measurement theory on admissible comparison.

Candidate distinctions were retained only when counterexamples showed that collapsing them produced false conclusions. The proposed framework was then tested against seven cases: identical implementations, distinct interfaces over one system, GUI versus CLI, API versus conversation, novice versus expert, accessibility variants, and simulated versus observed interaction. The framework is descriptive. It does not prescribe a single metric or ranking.

3. Why Similarity Is Not One Relation

Similarity is not a property that exists independently of representation and purpose. Tversky showed that judged similarity can be modeled through common and distinctive features, that feature salience is context-sensitive, and that similarity judgments can be asymmetric. Metric-space models impose identity, symmetry, and triangle inequality, but many psychological and practical judgments violate one or more of these assumptions.

Formal computer science reaches the same conclusion from another direction. Trace equivalence preserves possible observable action sequences. Bisimulation additionally preserves branching structure and stepwise matching. Testing equivalences preserve outcomes of specified tests. Observational equivalence depends on the contexts allowed to distinguish systems. The linear-time/branching-time spectrum demonstrates that there is no unique behavioral equivalence; relations are ordered by discriminating power.

Process mining separates model-to-model similarity, log-to-log similarity, and conformance between logs and models. Sequence comparison distinguishes edit distance, alignment, temporal warping, subsequence similarity, and probabilistic distance. Graph comparison distinguishes exact isomorphism, subgraph matching, edit distance, spectral similarity, and kernel-based similarity. These are not alternative implementations of one universal relation. They preserve different structures and answer different explanatory questions.

4. Canonical Distinctions

Interaction token identity concerns whether two references denote the same historically situated episode. It requires continuity under the declared participant, boundary, and timescale criteria. Copying a log does not create a second interaction token; replaying the same inputs does.

Interaction type identity concerns whether episodes instantiate the same declared type. Type membership depends on an abstraction function that suppresses some differences and preserves others. The same token can instantiate several types, such as authentication, error recovery, or turn-taking.

Equivalence is a binary relation of exact preservation under a declared comparison contract. It is never unqualified. Semantic equivalence may preserve denotation or task-relevant consequences. Trace equivalence preserves observable sequences. Bisimulation preserves branching response structure. Opportunity equivalence preserves declared action-availability relations. Outcome equivalence preserves criterion outcomes.

Similarity is graded proximity under declared dimensions, representations, alignment rules, metrics, weights, and tolerances. Similarity need not be symmetric when the comparison is directional, such as substitution, migration, or transfer.

Incomparability is not low similarity. It occurs when no defensible common comparison space exists: boundaries conflict, required variables are absent, scales are noncommensurate, observation processes differ without calibration, uncertainty overwhelms discrimination, or the proposed mapping changes the target itself.

5. Similarity Ontology

The minimal ontology contains:

Comparison target: the entity being compared, such as episode token, interaction type, protocol, opportunity profile, trace, behavior pattern, policy, strategy, outcome, or causal model.

Comparison frame: declared system boundary, participants, initial conditions, task, environment, timescale, and observation window.

Representation: the form in which each target is available for comparison.

Correspondence mapping: a relation aligning entities, states, actions, events, roles, or features across targets.

Invariant set: properties required to remain exact.

Variant set: properties permitted to differ.

Admissible transformation: a transformation treated as irrelevant for the comparison, such as renaming, temporal rescaling, stuttering, reordering of independent events, or modality substitution.

Observable set: distinctions available to the declared observer or instrument.

Distance or discrepancy: a typed measure of deviation.

Weighting and tolerance: importance assignments and thresholds for equivalence or practical similarity.

Uncertainty: error, missingness, model uncertainty, and confidence attached to inputs and results.

Validity domain: population, tasks, systems, contexts, and purposes over which the comparison claim is supported.

6. Typed Dimensions of Interaction Comparison

Semantic comparison asks whether operations, states, relations, or consequences denote the same declared target. It does not establish identical access, behavior, or outcomes.

Structural comparison concerns participants, coupling topology, state-transition structure, causal ordering, concurrency, branching, loops, and dependencies.

Protocol comparison concerns permitted messages or actions, sequencing rules, preconditions, responses, completion, and failure behavior.

Opportunity comparison concerns possible, capable, permitted, enabled, reachable, exposed, signaled, believed, reversible, and recoverable actions. Equality of backend actions does not establish opportunity equivalence.

Trace comparison concerns recorded or attributed event sequences or partial orders. It is observation-model dependent and may use edit operations, alignments, temporal normalization, or probabilistic distances.

Behavioral comparison concerns attributed trajectories or patterns, not raw records. It requires compatible segmentation and attribution rules.

Policy comparison concerns mappings from states or beliefs to action distributions. Strategy comparison concerns higher-level organization across contingencies and may remain latent even when policies are estimated.

Outcome comparison concerns declared success, utility, quality, safety, or other criteria. Equal outcomes do not imply equal process, opportunity, behavior, effort, or risk.

Causal comparison concerns whether the same dependency structure or mechanism generated observations. It cannot be inferred from trace similarity alone.

Measurement-relative comparison concerns whether two measured values are comparable under compatible instruments, scales, invariance assumptions, and uncertainty budgets.

7. Equivalence Hierarchy

The framework does not impose one total hierarchy because several dimensions are orthogonal. Within formal behavior, stronger relations generally imply weaker ones under fixed models: identity implies isomorphism under the chosen representation; isomorphism may imply bisimulation; bisimulation commonly implies trace equivalence; trace equivalence may imply equality of a selected outcome. The converses generally fail.

Across interaction dimensions, no universal implication holds. Semantic equivalence does not imply opportunity equivalence. Opportunity equivalence does not imply behavioral equivalence because agents differ. Behavioral equivalence does not imply causal equivalence. Outcome equivalence does not imply behavioral or interactional equivalence. Measurement equivalence is a precondition for empirical comparison, not a property of the interactions themselves.

The resulting architecture is a lattice of typed equivalence claims rather than a single ladder. A complete comparison report is therefore a vector of supported relations and distances, with explicit unknown or incomparable entries.

8. Formal Comparison Contract

Let interactions I and J be represented under contracts C_I and C_J. A comparison specification is:

K = <T, F, M, O, A, D, W, epsilon, U, V>

where T is the target type; F is the aligned comparison frame; M is the correspondence mapping; O is the observable set; A is the admissible transformation set; D is a family of typed distances or predicates; W is weighting; epsilon is the tolerance or equivalence bound; U is uncertainty; and V is the validity domain.

Exact equivalence on dimension k holds when, after an admissible transformation and valid correspondence, all required observables are preserved and d_k(I,J)=0 or satisfies the formal equality predicate. Approximate equivalence holds when the distance and its uncertainty interval remain within epsilon_k. Similarity is the reported distance profile rather than a categorical assertion. Incomparability holds when the mapping, frame, observation model, or scale conditions required by K cannot be justified.

A composite similarity score may be reported only when aggregation has a defensible measurement model. Weighted sums conceal compensation: severe inequality in permission or accessibility can be offset numerically by superficial trace similarity. The default output should therefore be a typed profile, not one scalar.

9. Evidence and Measurement Requirements

A comparison claim must identify its evidence level. Direct records include timestamps, messages, state snapshots, and instrument readings. Detected events are produced by segmentation or recognition rules. Attributed interaction and behavioral events add system or agent attribution. Policies, strategies, intentions, and causal mechanisms are inferred constructs.

Trace similarity is invalid when logging coverage differs materially. Behavioral similarity is invalid when segmentation or attribution rules differ without calibration. Group comparison is invalid when measurement invariance is untested. Cross-modality comparison is invalid when metrics encode modality-specific costs but are interpreted as general interaction cost.

Uncertainty should propagate through alignment, missing data, classification, metric estimation, and aggregation. A difference smaller than measurement uncertainty supports neither similarity nor dissimilarity. Reproducibility requires the target representations, mappings, code, parameterization, thresholds, exclusions, and sensitivity analyses to be available.

10. Comparative Cases

Identical implementations. Two episodes on the same software are not token-identical and need not be type-equivalent if initial state, authority, task, or participants differ. Implementation identity is weak evidence.

Different interfaces over one system. A GUI and CLI may be semantically equivalent at the operation layer while differing in exposure, signaling, sequence cost, recoverability, and trace distributions. Similarity must be reported by dimension.

API versus conversational interface. Both may invoke the same operation, but the conversation can introduce ambiguity, overclaim, confirmation structure, and nondeterministic interpretation. Backend outcome equivalence does not establish protocol or opportunity equivalence.

Novice versus expert. The projection and system-relative opportunity sets may be identical. Believed actions, strategies, search behavior, and recovery differ. The comparison target determines whether this is one interaction type with participant variation or different types.

Accessibility variants. Visual, auditory, and tactile projections can preserve semantics and completion while changing ordering, latency, memory burden, spatial comparison, and error recovery. Modality substitution is admissible only for explicitly preserved dimensions.

Simulated versus observed interaction. Simulation can reproduce traces or aggregate distributions without reproducing causal mechanisms, situated interpretation, or measurement error. Validation must distinguish trace fidelity, distributional fidelity, policy fidelity, and causal fidelity.

11. Admissible Transformations

Renaming is admissible for structural and protocol comparison when labels have a validated correspondence and no semantics are carried by the names.

Temporal translation is often admissible; temporal scaling is admissible only when absolute timing is not part of the target. Dynamic time warping may align patterns while concealing meaningful latency differences.

Stuttering or insertion of silent internal steps is admissible under weak behavioral equivalences when internal events are legitimately unobservable. It is not admissible when those steps affect effort, risk, energy, or recovery.

Reordering independent concurrent events is admissible under partial-order comparison when independence is established. Sequence metrics can falsely penalize equivalent concurrency.

Aggregation is admissible only when the coarse-graining preserves required invariants. Segmentation changes can manufacture similarity or difference.

Modality substitution is admissible for operation semantics under a valid cross-modal mapping, but not automatically for perceptual, temporal, behavioral, or accessibility comparison.

Implementation substitution is admissible when the comparison target is defined above implementation and equivalence obligations are verified. Identical output samples are insufficient.

12. Comparison Protocol

1. Declare whether the objects are episode tokens, types, models, traces, behaviors, opportunities, policies, strategies, outcomes, or mechanisms.
2. Establish compatible interaction boundaries, participants, tasks, initial conditions, and timescales.
3. Specify representations and observation models; audit missingness and logging coverage.

4. Define the correspondence mapping across states, actions, events, roles, and outcomes.
5. Select dimensions independently; do not infer one from another.
6. Declare invariants, admissible transformations, and prohibited transformations.
7. Choose metrics or formal predicates with scale assumptions and known failure modes.
8. Set tolerances before inspecting results where confirmatory claims are intended.
9. Propagate measurement and model uncertainty.
10. Report a typed comparison profile with equivalence, similarity, difference, unknown, or incomparable for each dimension.
11. Test robustness to alternative boundaries, mappings, metrics, weights, and segmentation.
12. State the validity domain and prohibited generalizations.

13. Falsification Tests

The theory should be narrowed or rejected if a single established equivalence relation can represent all relevant interaction comparisons without losing distinctions among semantics, protocol, opportunity, trace, behavior, policy, outcome, cause, and measurement. Current literature strongly contradicts that possibility.

The typed framework loses value if its dimensions cannot be measured reliably, if independent analysts cannot reproduce correspondence and segmentation decisions, or if typed profiles add no explanatory or predictive value beyond a well-chosen single metric.

The proposed incomparability category is falsified as practically necessary if every studied comparison can be defensibly embedded in a common space without arbitrary assumptions. Conversely, frequent analyst disagreement about mappings and boundaries would weaken claims of stable equivalence.

Claims of opportunity, behavioral, policy, or causal similarity must be rejected when based only on semantic or trace evidence. A composite score should be rejected when conclusions reverse under plausible weighting or when compensation masks a critical dimension.

14. Threats to Validity

The main threat is framework breadth. Typed comparison can become a checklist rather than a theory if dimensions are added without mechanisms or formal relations. The framework therefore limits itself to distinctions already supported across formal, behavioral, and measurement literatures.

Correspondence mappings can encode the conclusion. Analysts may align only convenient events or choose abstractions that manufacture equivalence. Pre-registration, independent coding, mapping audits, and sensitivity analysis are required.

Human interaction involves context, meaning, and evolving goals that cannot always be reduced to fixed feature vectors or transition systems. Strategy and causal similarity may remain underidentified. Cross-cultural and accessibility comparisons are especially vulnerable to treating instrument bias as participant difference.

Exact equivalence can be undecidable or computationally expensive for expressive models. Approximation introduces tolerance dependence. Empirical logs are finite samples and cannot prove equality of full behavior spaces.

15. Implications for Interaction Research

Similarity should not be treated as a single dependent variable. Research should ask which equivalence or distance relation is scientifically relevant to the claim. Interface migration may require semantic and opportunity preservation; accessibility evaluation may require outcome preservation without demanding trace identity; behavioral transfer may require type-level structural similarity while allowing different surface actions; formal verification may require bisimulation or refinement.

The framework also clarifies the next research stage. Transfer depends on a declared similarity relation between source and target interactions. Strategy research requires separation of trace similarity from latent organizational similarity. Formal-model selection requires identifying which equivalences the model can express and decide. Comparative interaction can then use typed profiles rather than ungrounded rankings.

16. Conclusion

The evidence supports a general theory of interaction comparison, but not a universal similarity metric. Interaction identity, equivalence, similarity, and incomparability are distinct. Every defensible claim is relative to a declared target, frame, representation, correspondence, observable set, admissible transformations, dimensions, metrics, tolerances, uncertainty model, and validity domain.

Existing literatures already supply the constituent formal and empirical tools. The needed contribution is a cross-disciplinary integration that prevents semantic, structural, opportunity, trace, behavioral, policy, outcome, causal, and measurement relations from being collapsed. The canonical output is a typed similarity profile. Scalar aggregation is optional and requires independent justification.

17. Reusable Artifacts

A. Similarity Ontology

Element	Definition
Comparison target	The entity or construct being compared.
Comparison frame	Boundary, participants, task, state, environment, and timescale.
Representation	The form available to the comparison procedure.
Correspondence mapping	Alignment across states, events, actions, roles, or features.
Invariant	Property required to remain exact.
Admissible transformation	Difference treated as irrelevant for a declared comparison.
Observable set	Distinctions available to an observer, instrument, or test context.
Distance/predicate	Typed rule used to determine discrepancy or equivalence.
Tolerance	Maximum admissible discrepancy for approximate equivalence.
Uncertainty	Error and indeterminacy in records, mappings, measures, and models.
Validity domain	Scope over which the comparison claim is supported.

Incomparability	Failure to establish a defensible common comparison space.

B. Similarity Taxonomy

Type	Preserved target	Required evidence
Token identity	Historical continuity of one bounded episode	Interaction contract, provenance
Type identity	Instantiation of the same abstraction-defined type	Classification rules
Semantic	Denotation, operation meaning, inferential consequences	Formal semantics, task contract
Structural	Topology, dependencies, branching, concurrency	Graphs, transition systems
Protocol	Sequencing, preconditions, responses, completion	Protocol/state-machine model
Opportunity	Feasible, permitted, exposed, signaled, believed, recoverable actions	Opportunity contract
Trace	Recorded or attributed event order and timing	Trace model and observation contract
Behavioral	Attributed trajectory or pattern	Behavioral evidence contract
Policy	State/belief to action distribution	Policy estimates or model
Strategy	Higher-order organization across contingencies	Multi-episode evidence
Outcome	Declared criterion results	Outcome contract
Causal	Generating dependency or mechanism	Intervention/causal evidence
Measurement-relative	Comparability of measured quantities	Measurement invariance and uncertainty

C. Interaction Equivalence Hierarchy

Relation	Required sameness	Observation basis
Token identity	Same historical episode	Boundary continuity and provenance
Isomorphism	Same structure up to bijective renaming	Exact structural mapping
Bisimulation	Stepwise matching including branching	Transition relation
Trace equivalence	Same observable traces	Trace language
Testing/observational equivalence	No allowed test/context distinguishes systems	Declared test contexts
Semantic equivalence	Same denotation or declared semantic consequences	Semantic contract
Opportunity equivalence	Same typed opportunity profile	Opportunity dimensions
Behavioral equivalence	Same attributed behavior under observation contract	Behavioral variables
Policy equivalence	Same action distributions under aligned states/beliefs	Policy mapping
Outcome equivalence	Same declared criterion distribution	Outcome metric
Approximate equivalence	Typed distance within tolerance and uncertainty	Metric plus epsilon
Incomparability	No justified common space	Failed alignment/validity conditions

D. Similarity Contract

- Comparison identifier and purpose
- Target types and token/type status
- Interaction contracts and aligned boundaries
- Participants, tasks, environments, initial conditions, and timescales
- Representations and observation models
- Correspondence mapping and mapping evidence
- Required invariants and permitted variants
- Admissible and prohibited transformations

- Comparison dimensions
- Formal predicates, metrics, scales, and parameters
- Weights and aggregation model, if any
- Equivalence tolerances and decision rules
- Missingness and uncertainty budget
- Sensitivity analyses
- Evidence level for each claim
- Validity domain and prohibited generalizations
- Final typed profile: equivalent, approximately equivalent, different, unknown, or incomparable by dimension

E. Similarity Comparison Protocol

1. Declare whether the objects are episode tokens, types, models, traces, behaviors, opportunities, policies, strategies, outcomes, or mechanisms.

2. Establish compatible interaction boundaries, participants, tasks, initial conditions, and timescales.

3. Specify representations and observation models; audit missingness and logging coverage.

4. Define the correspondence mapping across states, actions, events, roles, and outcomes.

5. Select dimensions independently; do not infer one from another.

6. Declare invariants, admissible transformations, and prohibited transformations.

7. Choose metrics or formal predicates with scale assumptions and known failure modes.

8. Set tolerances before inspecting results where confirmatory claims are intended.

9. Propagate measurement and model uncertainty.

10. Report a typed comparison profile with equivalence, similarity, difference, unknown, or incomparable for each dimension.

11. Test robustness to alternative boundaries, mappings, metrics, weights, and segmentation.

12. State the validity domain and prohibited generalizations.

F. Admissible Transformation Matrix

Transformation	Candidate dimensions	Admissible?	Principal constraint
Label renaming	Structural/protocol	Yes, with bijection	No if labels carry semantic or signaling content
Time translation	Trace/behavior	Usually	No when calendar position matters
Time scaling	Pattern/strategy	Conditional	No when absolute latency is a target
Dynamic time warping	Temporal pattern	Conditional	Can conceal meaningful delay
Silent-step insertion	Weak behavioral equivalence	Conditional	No when internal work affects cost/risk
Independent-event reordering	Partial-order behavior	Yes if independence proven	No if order has causal/semantic force
Aggregation/coarse-graining	Type/structural	Conditional	Must preserve declared

			invariants
Modality substitution	Semantic/outcome	Conditional	Not automatically perceptual/behavioral equivalent
Implementation substitution	Semantic/protocol	Conditional	Requires verified obligations
Participant substitution	Type-level comparison	Conditional	Not token identity; observer assumptions change

G. Falsification Matrix

Claim/test	Falsifying evidence	Consequence
F1 Universal relation	One established relation captures all comparison dimensions without loss	Remove typed framework and adopt it
F2 Unreliable dimensions	Opportunity, behavior, or strategy dimensions cannot be measured reproducibly	Restrict framework to reliable dimensions
F3 No added value	Typed profile gives no explanatory/predictive gain over one metric	Prefer the simpler metric
F4 Arbitrary mappings	Independent analysts cannot agree on correspondence under explicit rules	Treat claims as analyst-relative or incomparable
F5 Incomparability unnecessary	All cases admit justified common spaces	Remove incomparability as a primitive status
F6 Aggregation robustness	Composite conclusions reverse under plausible weights	Reject scalar summary
F7 Trace sufficiency	Trace similarity predicts every relevant comparison outcome	Collapse higher dimensions where demonstrated
F8 Cross-context invariance failure	Measurement relations do not remain invariant across groups/modalities	Limit validity domain

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